

Paper 2.25 - Toolkit for the Correction Surface

CQE-paper-02.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-13

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-02.25.md

Paper 2.25 - Toolkit for the Correction Surface

Purpose

Paper 2.25 describes the tools for reviewing Paper 2. These tools expose the same finite correction theorem by script, table, or hand reconstruction. They do not add claims beyond Paper 2.

Review Objects

The toolkit uses five objects:

```
LCR state      = (L,C,R)
Rule30 update  = L xor (C or R)
Rule90 update  = L xor R
correction     = C and not R
D4 projection  = axis/sheet label for the state
```

Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-02/verify_correction_surface.py
production/formal-papers/CQE-paper-02/correction_surface_receipt.json
```

The verifier enumerates $\{0,1\}^3$, computes Rule 30, Rule 90, and the correction term, then checks that:

```
Rule30 = Rule90 xor correction
correction fires only on (0,1,0) and (1,1,0)
the firing D4 coordinates are (2,0) and (3,1)
nonzero residue is marked as obligation, not proof
```

Analog Toolkit

A physical reconstruction requires only:

- three marked positions,
- binary tokens or marks,
- one table row for Rule 30,
- one table row for Rule 90,
- one residue mark,
- one receipt line.

Procedure:

```
draw L | C | R
place binary values
compute Rule30 = L xor (C or R)
compute Rule90 = L xor R
compute corr = C and not R
check Rule30 = Rule90 xor corr
if corr = 1, write an obligation row
```

The analog route should produce exactly two residue rows:

```
(0,1,0)
```

(1, 1, 0)

Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

Which two binary LCR states make C and not R fire?
Does a nonzero correction close the failed route as proof?
What D4 coordinates are exported by the correction states?

Expected answers:

(0, 1, 0) and (1, 1, 0)
no
(2, 0) and (3, 1)

Boundary

Paper 2.25 is a toolkit supplement. If tool use uncovers a new correction, projection, or route, it must be promoted through Paper 2.50's validation contract before it can change the scientific paper.